

P3699R0R1

Rename ``conqueue_errc``

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Abstract

This paper proposes to rename `std::conqueue_errc`.

Table of Contents

1 Revision History

2 Introduction

3 Name Options

4 Proposal

References

Informative References

§ 1. Revision History

This paper revises P3699R0 - 2025-05-17 as follows:

- adding more name options

§ 2. Introduction

The proposal for concurrent queues [C++ Concurrent Queues \(P0260\)](#) uses an enumeration to specify possible return states of some queue operations. This enumeration is called `conqueue_errc` as it was originally meant for exceptions. However, the concurrent queue proposal doesn't propose it's own exception class anymore, so the name is misleading.

```
enum class conqueue_errc { success, empty, full, closed, busy, busy_async };
```

§ 3. Name Options

In initial revisions of P0260 the enumeration was called `queue_op_state`. But `op_state` has a specific meaning in S/R context that completely different.

Dropping the `op` gives us `queue_state`. To make it clear that it's specific to concurrent queues we could also name it `conqueue_state`.

From the reflector we got more proposals, so here's a list:

- `queue_state`
- `conqueue_state`
- `queue_status`
- `conqueue_status`
- `conqueue_result`
- `conqueue_op_status`
- `conqueue_op_result`

§ 4. Proposal

Rename `std::conqueue_errc` to `std::XYZ`.

§ References

§ Informative References

[P0260]

C++ Concurrent Queues (P0260). URL: <https://wg21.link/P0260>